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# Robot-supported multiplayer rehabilitation: Feasibility study of haptically linked patient-spouse training

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*Abstract*— **Multiplayer environments are thought to increase and prolongate active participation in robot-aided rehabilitation. We expect that environments linking patients with their spouses will particularly foster active participation. Thus, we developed two multiplayer games to link the game experience of two players: an Air Hockey game and a Haptic Kitchen game. In the competitive Air Hockey game, differences in skill levels between players were balanced by individualizing haptic guidance or damping forces. In the Haptic Kitchen game, a healthy player could support the patient's movements using a virtual force field. The two players could control the haptic interaction since both the force field and the point of application were visualized. We tested the haptic performance balancing algorithm of the Air Hockey game and the spouse-controlled haptic support of the Kitchen game with patients post-stroke who trained both single- (i.e., alone) and multiplayer training (i.e., with spouse) in eight therapy sessions lasting 45 min each. Mean total rating in Intrinsic Motivation Inventory was 46.9 points (out of 63 points) for multiplayer modes, and 42.7 points for single player modes, respectively. The spouses applied the haptic support in the Haptic Kitchen game during 42% of the total game duration. We are currently testing more patient-spouse couples to better understand the effects of using these haptic approaches on the behavior and recovery of patients. We foresee this approach can improve the motivation during training and positively influence the at-home behavior of patients, an important goal of rehabilitation training efforts.**

## I. INTRODUCTION

Robots are increasingly applied in motor therapy during neurorehabilitation. As a tool for training, robots have the potential to provide high-intensity training – in the range of hundreds of movements per hour. Robot-based training, however, lacks the social interaction that takes place in conventional therapy when a patient interacts with a therapist or with other people which may serve for reassurance and pressure to perform well [1]. Multiplayer settings can foster social interaction. Indeed, in neuromuscular training after stroke, motivation of the subjects was increased by multiplayer modes [2], [3], [4].

However, multiplayer modes per se do not affect the enjoyment of the patient [2]. When playing with a friend or relative, verbal interaction was increased being an indicator for increased enjoyment. Furthermore, based on the flow theory, we could expect that balanced competitive games (i.e. similar success rate for each player) also increase the motivation [5]. In multiplayer studies, changes in game mechanics (e.g., increased avatar sizes for the less skilled player) were used to balance out different skill levels [2]. Such balancing strategies may visually reveal the different skill levels of the players which can be embarrassing for the worse performing player – i.e. usually the patient. Furthermore, since not every game input device suits for every player's skill, game input devices for patients and game input devices for healthy players need to be jointly integrated in multiplayer games.

We implemented and successfully tested an algorithm that controlled a virtual damping coefficient in an arm exoskeleton joint to balance game performance of healthy players [6]. However, a modulation of the damping coefficient is not suitable to manipulate the game experience of neuromuscular patients: Instead of damping forces, patients often need supportive forces to reach a desired game performance for a meaningful multiplayer gameplay. If the game performance of a patient and of a co-player could be controlled individually, a balanced multiplayer game could be achieved. Up to now, such a haptic performance balancing controller was not integrated in multiplayer therapy gaming.

Besides entertaining games, such as single- or multiplayer competitive games, virtual environments for training of activities of daily living (ADL) are integrated in single player neuromuscular therapy after stroke [7]. Although the context of ADL environments might increase verbal interaction in multiplayer gaming when playing with someone sharing daily life, e.g. spouse, multiplayer modes are not integrated in robot-assisted devices yet [8]. By using a haptic input device for one or two of the players, haptic interaction complements verbal interaction and therefore may boost effects of social interaction when jointly experiencing the ADL environment.

We recently developed a novel multiplayer setting that facilitates haptic performance balancing in a multiplayer mode of a competitive game and haptic support controlled by a healthy co-player in a cooperative ADL environment. In the here presented study we aimed to test with stroke patients and their spouses if such multiplayer modes result in a higher motivation compared to single player modes. Furthermore, we aimed to test if haptic performance balancing decreases the discrepancy between patient's success rate and predefined desired success rate during gameplay, and if haptic support is applied by the spouse players.

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## II. METHODS AND MATERIALS

### A. Experimental setup

Two devices were selected for the study (Figure 1):

- For the patient: the ARMin arm rehabilitation robot for patients with neurological injuries (4th generation, [9], [10])..
- For the spouse: the HTC Vive's handheld (High Tech Computer Corporation, Taoyuan, Taiwan).

A monitor, placed $1.2\,\mathrm{m}$ from the player's shoulders, displayed the virtual reality environments. An occupational therapist monitored the players and devices during training.



Fig. 1: Multiplayer setup with haptic interaction. The patient played with the ARMin robot (center) while the spouse played with the HTC vive handheld (left). The monitor displayed the virtual reality environment to both players. A therapist (right) guided and observed the therapy session.

### B. Virtual reality environments

We created two virtual reality environments with haptic elements for multiplayer interaction: an Air Hockey game and the Haptic Kitchen game (Figure 2).

*1) Air Hockey game:* The goal in the Air Hockey game was to defend one's own goal (defensive task) while scoring on the opponent's side (offensive task). For the ARMin, single-joint movements (e.g. elbow flexion/extension) were mapped to the mallet's position along the horizontal axis. From the game's dynamics, a player is required to do time-constrained reaching movements to meet the game's goal.

We extended an existing controller, designed to adapt the game's difficulty by damping the player's movement [6], with a controller that can support the player. The support was needed for patients who otherwise would be unable to play the game competitively against an unimpaired player. The therapist was free to select the joint that was to be used by the patient before starting a round of the game. All other joints were fixed in place by a position controller.

In single player mode, the opponent's mallet was controlled by the computer using minimum jerk trajectories based on assessments with unimpaired participants [6], [11]. In multiplayer mode, the opponent's mallet was controlled by the spouse by moving the HTC Vive hand held controller. The spouse's movement was mapped from a $0.6\,\mathrm{m}$ range ($-0.3\,\mathrm{m}...0.3\,\mathrm{m}$) in the lateral direction with the origin at the spouse's shoulder. Movements in the vertical or anterior-posterior direction did not affect the mallet's position. In the game environment, side walls constrained the position of the puck and the mallets.

To balance gameplay, we focused on the defensive task of the game. The defensive task required the players to reach the approaching puck within a given time window. This embeds training of time-constrained reaching movements in a gamified environment. Success in this task was to reach the puck in time; failure was allowing the puck to pass by and reach the players goal zone. We defined the success rate at defending as the number of successful hits divided by the number of total defensive trials. We targeted a success rate, $p_\infty$, of $70\,\%$ for every player. For the patient we adapted, after each trial, the level $\beta_i$, of damping or support of the movement. The adaptive algorithm used is described in detail in [6]. The $\beta_i$ coefficient translated into forces that ranged from zero guidance ($0\,\%$) to full support ($100\,\%$). For a $\beta_i$ coefficient below 0, the forces turned from supportive to resistive. The initial level, $\beta_0$, was set by the therapist and constrained to only supportive forces.

For a support coefficient $\beta_i$ between $0\,\%$ and $100\,\%$ the patient was guided haptically along a minimum jerk trajectory with an error reducing PD controller. The stiffness of the haptic guidance increased proportionally to the support coefficient $\beta_i$. For support coefficients below $0\,\%$ the movement was damped by forces that increased proportionally to a decreasing $\beta_i$.

$$Zi = \begin{cases} support, & \text{if } \beta_i \geq 0 \\ damping & \text{if } \beta_i < 0; \end{cases} \tag{1}$$

$Zi = support$: The support force reduced the reaching errors of the patient and thus increased the ability to successfully perform the reaching task.

$$e_{q,j}(t) = q_{c,j}(t) - q_j(t) \tag{2}$$

$$\dot{e}_{q,j}(t) = \dot{q}_{c,j}(t) - \dot{q}_j(t) \tag{3}$$

$$F_{i,j}(t) = -\beta_i \left(k_{p,j} e_{q,j}(t) + k_{d,j} \dot{e}_{q,j}(t)\right) \tag{4}$$

Joint $j$ was selected by the therapist, and $k_{p,j}$ and $k_{d,j}$ were predefined control parameters. The angular position $q_j$ of joint $j$ was controlled by the patient. The trajectory $q_{c,j}(t)$ was calculated for every defensive task as soon as the puck collided with the opponent's mallet. The trajectory had a reaction time component and a minimum jerk component. We used Fitt's law to estimate a reaction time of about $0.3\,\mathrm{s}$ from previous study data [6]. From the start position $q_{c,j}(t_0)$ to the target position $q_{c,j}(t_c)$ the trajectory was planned for a minimum jerk movement for the player's mallet. The start position $q_{c,j}(t_0)$ was the current position of the player's mallet when the puck collides on the opponent's side. The target position $q_{c,j}(t_c)$ was the position for a collision with the center of the puck. The time of collision $t_c$ was known since the puck's trajectory had known direction and speed. A time band and position band around $t_c$ and $q_{c,j}(t_c)$ offered different locations for a successful collision.



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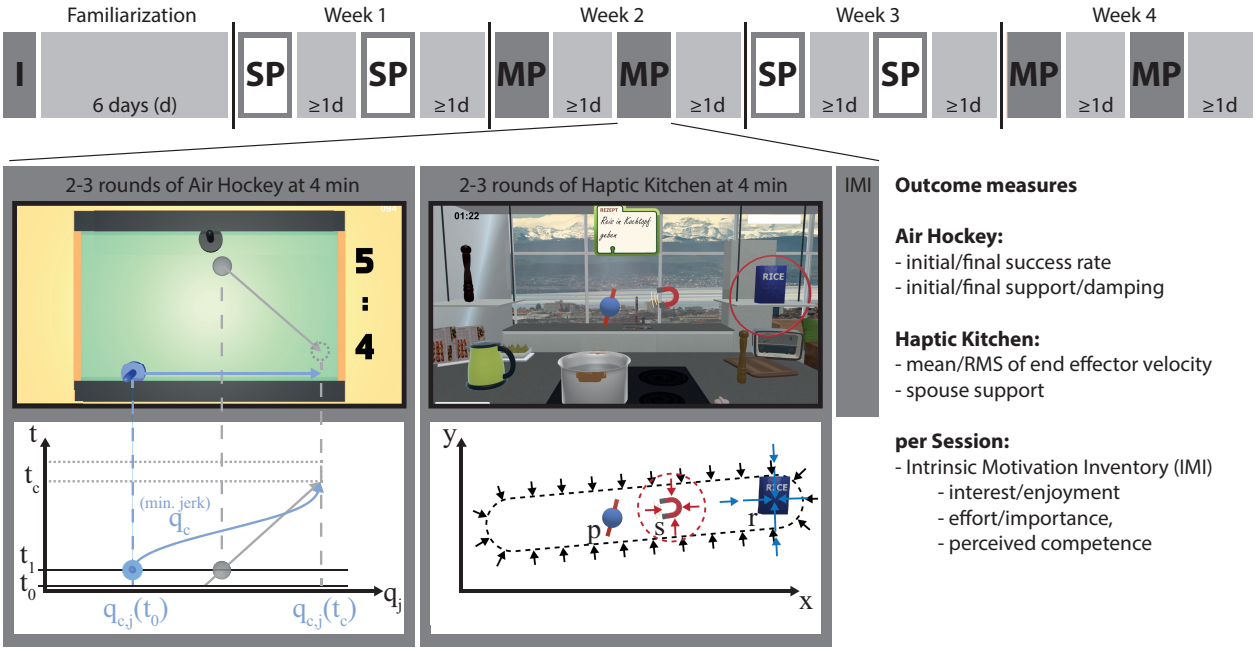



Fig. 2: Study protocol. In day one of the study, patients were instructed (I) and their active range of motion was assessed. After one week of familiarization of the therapist-instructed at-home tasks, participants engaged in four weeks of training with two training sessions per week. Week one and three of the intervention sessions were single player sessions (SP). Week two and four of the intervention sessions were multiplayer sessions (MP). In each session, 2-3 rounds of the Air Hockey game were played and 2-3 rounds of the Haptic Kitchen game were played. Each round lasted about four minutes. Bottom-left: From the known Air Hockey game mechanics, the trajectory of the puck is known (gray) and used to define a time-constrained minimum jerk trajectory $q_{c,j}$ for the patient's mallet (blue) from the starting point at $t_0$ to the end of the trajectory at $t_c$. Bottom-center: In the Haptic Kitchen game, three forces fields were used: 1) a path controller haptically supported the patient at the end effector $p$, 2) a force field around the target position $r$ attracted the end effector, $p$ when it was close by, and 3) for the multiplayer condition, a magnet-like force field around the spouse's end effector, $s$, allowed the spouse to guide, or restrict, the patient's movements by a virtual connection to the ARMin robot.

$Z_i = damping$: The damping force limited the maximum velocity that a player can reach and thus affected the ability to defend his or her goal [12].

$$\mu_{i,j} = \beta_i \cdot d_j \text{ Nms/}^\circ \tag{5}$$

$$F_i, j(t) = \mu_{i,j}\dot{q}(t) \tag{6}$$

The damping factor $d_j$ was pre-defined for each joint $j$. The damping coefficient $\mu_{i,j}$ was defined as the damping factor $d_j$ multiplied with the current support coefficient $\beta_i$.

The adaptation of support, or resistance, for non-haptic interfaces was done virtually. For the computer player we iteratively adapted the maximum velocity of the mallet (i.e. simulating a damping force). For the spouse, we iteratively adapted the mapping between the handheld controller and the mallet's position on the screen. The mapping ranged from assistive to disruptive using a mapping coefficient $\beta_{s,i}$. In the assitive scenario ($\beta_{s,i} < 25$), the mapping reduced the position errors between a minimum jerk trajectory and the spouse's movements. The reduction of errors was maximum when $\beta_{s,i} = 0\,\%$. In the disruptive scenario ($\beta_{s,i} \geq 25$), the mapping was visually distorted by increasing the discrepancy between the mapped player's movement and the minimum jerk trajectory. For a $\beta_{s,i}$ coefficient value of 25 the mapping was unchanged. The therapist set the initial mapping coefficient $\beta_{s,0}$ between $0\,\%$ and $100\,\%$.

*2) Haptic Kitchen game:* Although both the ARMin robot and the HTC Vive are capable of mapping three dimensional movements, in the Haptic Kitchen game we only considered movements in a two-dimensional plane. This plane was defined by a coordinate system $(x, y)$. From the patient's point of view, the $x$ coordinate pointed in rightward direction and $y$ coordinate in upward direction. For the ARMin, the origin of the coordinate system was situated in an assumed fixed shoulder joint of the patient. For the HTC Vive, the origin of the coordinate system was situated with outstretched arm and a total range of $0.6\,\mathrm{m}$ in horizontal direction (left-right) and $0.4\,\mathrm{m}$ in vertical direction (up-down). The anterior-posterior direction was integrated as a free degree-of-freedom not affecting the players' avatar position for both players. End effector forces from the robot were only applied in $x$ and $y$ direction.

The Haptic Kitchen game was designed to train an activity of daily life. The new game consisted of several recipes that were split into a number of actions that the patient had to fulfill in series. The kitchen table and weight of the objects were haptically rendered. All objects' positions, i.e. target position, were within the active range of motion (aROM) of the patient. The aROM of the patient was assessed in the first day of the study.

In single player mode the Haptic Kitchen game supported the patient with an assist-as-needed task space path controller [13]. A haptic tunnel limited the movements perpendicular to the direct path of the current position of the patient's hand to the target position. A haptic wall, moving with a five second delay from the start of the trajectory and moving along the path to the target, guided the patient's hand to reach the target position within 13.5 seconds. A third force



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field, located near the target position, attracted the patient's hand to the target position when the end effector was close by.

$$\boldsymbol{F_r}(F_x, F_y) = f(\boldsymbol{p}(p_x, p_y), \boldsymbol{r}(r_x, r_y)) \tag{7}$$

$$\boldsymbol{e_r} = \boldsymbol{r} - \boldsymbol{p} \tag{8}$$

$$\boldsymbol{F_r} = \left(\exp\left(-0.2\sqrt{e_{r,x}^2 + e_{r,y}^2}\right)\right) \cdot \left(\frac{\boldsymbol{e_r}}{|\boldsymbol{e_r}|} - 0.1 \cdot \dot{\boldsymbol{p}}\right) \tag{9}$$

$F_r$ were the virtual forces applied by the robot at the end effector based on the current target's position. $p$ was the current end effector position and $r$ was the target position.

The exponential function was selected to have a moderate attraction when a target was far and a strong attraction when the end effector was close to the target. This force field requires the patient to apply forces to reach the target while helping stabilize the end effector when it is close to the target to grasp the object. The therapist could choose the amount of support, i.e. stiffness of haptic tunnel and haptic wall and width of haptic tunnel, before starting the game.

In multiplayer mode the assist-as-needed controller was adapted to enable haptic interaction with the spouse. The path control forces remained, but the moving haptic wall guiding the patient towards the target in single player mode was removed. Instead, the spouse's movements were coupled to the patient's hand by a magnet-like force field similar to the one applied at the target (see Equations 7 to 9). The spouse could enable the force field by pulling the handheld's index trigger. Releasing the trigger disabled the force field again.

$$\boldsymbol{F_s}(F_x, F_y) = f(\mathbf{p}(p_x, p_y), \mathbf{s}(s_x, s_y)) \tag{10}$$

$$\boldsymbol{e_s} = \boldsymbol{s} - \boldsymbol{p} \tag{11}$$

$$\boldsymbol{F_s} = \max\left(0, \left(0.2 - 0.005\sqrt{e_x^2 + e_y^2}\right)\right) \cdot \left(\frac{\boldsymbol{e_s}}{|\boldsymbol{e_s}|} - 0.1 \cdot \dot{\boldsymbol{p}}\right) \tag{12}$$

$F_s$ were the virtual forces applied by the robot at the end effector based on the hand position of the spouse. These forces were translated into motor torques by the Jacobian. $p$ is the current end effector position and $s$ is the hand position of the spouse.

Instead of using an exponential function, a linear function was used as it was more intuitive for the spouse. The influence of the force field was restricted to a distance of $0.4\,\mathrm{m}$.

### C. Study protocol

The study was approved by the responsible ethical institution (KEK-ZH-Nr. 2015-0013, Zurich, Switzerland) and by clinicaltrials.gov (NCT02720341).

In eight sessions (four single player and four multiplayer sessions) couples (stroke patient and their spouse) tested the two games using an A-B-A-B experimental design (Figure 2). A couple was included if the patient suffered from decreased motor function in one or two arms due to a stroke and met the inclusion criteria (aged at least 18 years, no excessive spasticity of the affected arm represented by modified Ashworth Scale greater than 3, no serious medical or psychiatric disorders assessed by the physician, no orthopaedic, rheumatological, or other disease restricting movements of the paretic arm, no shoulder subluxation, no skin ulcerations at the paretic arm, ability to communicate effectively with the examiner such that the validity of the patient data could not be compromised, no cybersickness, no serious cognitive defects or aphasia preventing effective use of ARMin).

Patients did both, single player robot-assisted therapy (A) and multiplayer robot-assisted therapy (B). One week after a familiarization session, where active range of motion was assessed and an at-home task for the patients was instructed, four weeks of training each were performed. The first and third week were single player sessions; second and fourth week were multiplayer sessions. Every session consisted of a standard $45\,\mathrm{min}$ therapy unit carried out by an occupational therapist. The session was designed to include at least two $4\,\mathrm{min}$ games of the Air Hockey game and at least two $4\,\mathrm{min}$ games of the Kitchen game. Every game was followed by a break of at least $1\,\mathrm{min}$. The therapist was free to select the joints and initial support for the Air Hockey game, and the support level for the Haptic Kitchen game. The patient and spouse were free to interact during the multiplayer sessions.

In the multiplayer sessions, patients trained in a team with their spouse. The therapist was free to select the initial difficulty level of the spouse. The spouses played with their dominant arm.

To measure the technical feasibility of the intervention, we monitored the patient's success rate and the level of support or damping of the movement in the Air Hockey game. In the Haptic Kitchen we assessed the patient's velocity profile by extracting the mean and the root-mean-square values. We also assessed the amount of time the spouse triggered the haptic support.

After each session the patients rated three statements from each of the three subscales interest/enjoyment, effort/importance, and perceived competence of the Intrinsic Motivation Inventory (IMI) [14]. Each statement had to be rated on a 7-point scale, with 1 indicating not at all true, and 7 indicating very true. From the ratings we calculated sums for all subscales and a total sum of the IMI ratings.

## III. RESULTS

Two chronic stroke patients (patient 1: female, 66 years old, two strokes, 12 and 27 months ago, right arm affected; patient 2: male, 34 years old, several strokes in the past 8 years, left arm affected) participated in the study, each with his/her spouse. None of the sessions was interrupted due to technical or motivational issues. All players reported that the games were fun to play and, in multiplayer settings, the players highly valued the new therapy experience. None of the patients reported any pain. Both spouses noted that the game was exhausting for them.



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The haptic balancing of the Air Hockey game used both support and additional damping (Figure 3A). The success rate in the Air Hockey game was calculated with a moving window of 25 trials. In the Haptic Kitchen game the spouse used the force field in phases (Figure 3B).

Player 1 played the Air Hockey game 23 times (thirteen times with forearm supination/pronation, nine times with elbow flexion/extension, and once with horizontal shoulder ab-/adduction). The mean of all session's success rate of patient 1 changed from initial 73 % to final 74 % in single player and from initial 52 % to final 70 % in multiplayer mode. Player 2 played the Air Hockey game 22 times (thirteen times with elbow flexion/extension and nine times with horizontal shoulder ab-/adduction). For player 2 the success rate remained unchanged from initial 79 % in single player and decreased from initial 83 % to final 78 % in multiplayer mode (Figure 4A). The final support was higher for every game in multiplayer conditions compared to what the therapist set pre-game. Two sessions of patient 2 (one single player and one multiplayer) had corrupt data from the Kitchen game which was excluded from the result plots (Figure 4B). The sum of the IMI ratings ranged from 30 to 57 points for single player and from 34 to 58 points for multiplayer sessions out of 63 possible points (Figure 4C).

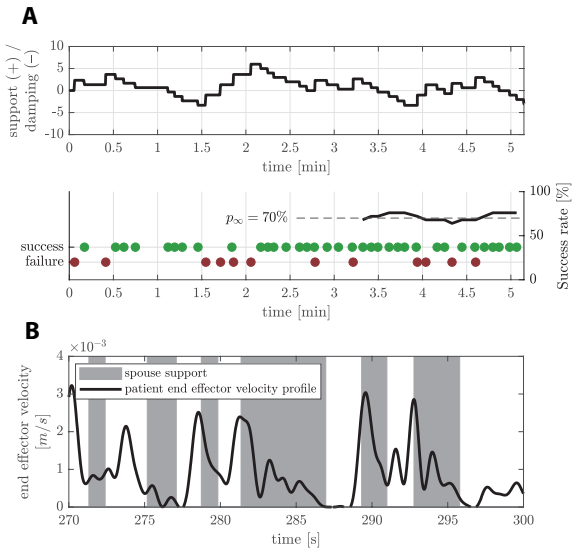



Fig. 3: Exemplary data of the two games. A: Air Hockey game data. Support and damping values from a single player session of patient 1. The success (green dots) and failure (red dots) in defending trials were causing a change in the support. The success rate (solid line) is plotted against the desired success rate $p_\infty = 70\,\%$ (dashed line). B: Kitchen game data. The velocity profile of the end effector of the second multiplayer session of patient 2 was smoothed for visualization using a fourth-order butterworth filter (cut-off frequency: 1 Hz). Phases where the spouse triggered the force field are indicated (gray).

## IV. DISCUSSION

The combination of haptic and non-haptic approaches to balance skill levels in the Air Hockey game allowed for competitive gameplay for both players, i.e., patient and spouse. By using a haptic approach with the patient, the modulation of the game conditions was not apparent to the patient. The haptic approach complements existing methods where the

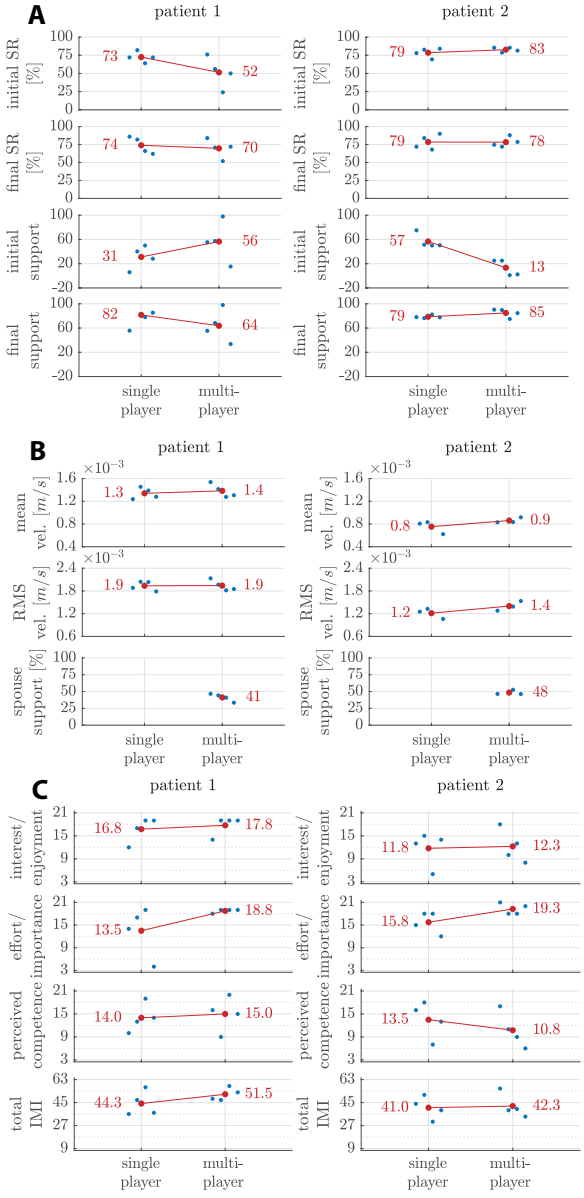



Fig. 4: Summary of the results. The mean results of the sessions of each patient in clouds (blue dots) and the mean for each training mode (red dots) are reported. A: Air Hockey results of games with at least 25 defensive trials. The initial and final success rates (SR) are the mean values over one session for the first and last calculated success rates. B:Mean and RMS values of velocity during the Kitchen game. Spouse support is presented relative to the total game time and only for multiplayer sessions. C: Intrinsic Motivation Inventory (IMI) scores. The self-assessed sub-scales interest/enjoyment, effort/importance, and perceived competence of the IMI are presented. The sum of all sub-scales is presented in the total IMI sub-figure.

difficulty is adjusted non-haptically only by changing avatar size or ball speed [2]. The non-haptic manipulation of errors to adapt skill levels also conceals different skill levels. The algorithm that was used to adapt the condition for each player is therefore applicable in both haptic and non-haptic devices.

Compared to a previous publication [6] we facilitated a within-game transition from conditions of patient support to patient challenge. In single sessions the algorithm seemed to work as the algorithm tends to adapt the condition towards desired success rate in particular when the initial condition was wrongly selected by the therapist (e.g. adapting from an



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initial success rate of $25\,\%$ to $52\,\%$. However, the game time of $4\,\mathrm{min}$ and the selection by the therapist of the joint to be trained impeded full balancing of performance within game duration or transition of conditions between games.

In the Haptic Kitchen game we offered the opportunity to the spouses to use a force field to support, or hinder, the impaired partner in a virtual activity of daily living. In nearly half of the game time ($41\,\%$ by the spouse of patient 1, and $48\,\%$ by the spouse of patient 2) the opportunity was used to haptically support the impaired partner.

In both games we noted an increased amount of verbal interaction compared to a patient-patient setting in another study we performed. In particular, the kitchen environment provided context for verbal interaction of the couples, including mentioning about their current at-home behavior.

We expect that including a close relative or spouse into the therapy process will positively affect the patients' at-home behavior. Changing a health-relevant behavior, such as adhering to therapy recommendations or changing one's diet is dependent on several self-regulatory skills as, for example, specified in the Health Action Process Approach [15], [16]. Stroke patients with arm impairments need to change their at-home behavior to support the occupational and physical training plan. To promote and maintain therapy progress, homework tasks designed by therapists need to be repeatedly trained at home and the impaired arm needs to be involved in activities of daily living. However, good intentions are often not enough for a successfully maintained behavior change [17]. The social environment, e.g. the spouse, of the patient may facilitate (or hinder) this change in behavior. Social exchange processes, such as social support might play a prominent role in health behavior change [18]. We expect that the multiplayer sessions promote positive social exchanges in spouses and thereby foster better adherence to therapy recommendations. The ultimate goal is to increase the use of the impaired arm in the patient's everyday life. The health behavior change of the patients involved in the present study was monitored by wrist-worn accelerometry and related motivational, volitional and social variables self-reports. These data will be reported in future publications.

## V. CONCLUSIONS

Rehabilitation robots in multiplayer games have the potential to haptically support patients in competitive games and facilitate haptic interaction in cooperative games. In competitive games, haptic guidance and damping forces can be used to offer difficulty adaptation for involved players. In cooperative games, haptic support can be used by the patient's spouse to haptically interact with their partner. Based on the promising results we found with these two couples measured so far, we will continue to investigate the effect of multiplayer games using haptic therapy devices on patient's kinematics and game experience. Our long-term goal continues to be the development of training environments that positively influence the patient's at-home behavior.


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